Toward a four-dimensional Yang–Mills construction

Notebook 03: current commutators and Gaussian cutoff clusters

Research working document

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Abstract

Several concrete estimates are established. First, the covariant cutoff commutator in the deposited current argument is bounded using only first differences, with its advertised dimension-dependent constant. Second, sharp large-field defects of a uniformly accretive complex Gaussian admit a convergent hard-core polymer expansion and a uniformly small logarithm in an anchored, exponentially weighted norm. The second proof differentiates the Gaussian covariance, never the sharp indicator. The cluster bound also gives a direct two-source Hessian estimate and a thermodynamic-limit criterion for the Gaussian cutoff contribution to the quadratic coefficient. These estimates are uniform in volume under their stated hypotheses. The unresolved interfaces are explicit: control of the actual combined current at complex decoupling parameters, and strict local dependence of cutoff activities on the Yang–Mills background. No continuum construction or mass-gap result follows without those and the other inputs listed in Notebook 01.

Status and scope. The proofs below have been checked internally, including independent checks of the commutator algebra and the Gaussian tree summation. They have not undergone external mathematical review. The standard forest formula and polymer convergence theorem are cited explicitly; no novelty is claimed for them or for these applications. References to Raw I concern commit `e392ef4501 3a581664bab76b90867170b30fe10e` of the deposited account.

1 A cutoff commutator needing only first differences

Fix lattice spacing $\eta > 0$. For a complex covariant transport define

$$(T_\mu B)(x) = R_{x,\mu} B(x + \eta e_\mu), \qquad (T_\mu^{-1} B)(x) = R^{-1}_{x-\eta e_\mu,\mu} B(x - \eta e_\mu),$$

$$\nabla_\mu = (T_\mu - I)/\eta, \qquad \nabla_\mu^* = (T_\mu^{-1} - I)/\eta.$$

Here star is the bilinear transpose of the difference operator. After complexification it need not be its Hermitian adjoint. Suppose the transports and their inverses on the relevant stencil have norm at most τ. For a scalar cutoff χ, let s_1 bound all its first differences and s_2 all its mixed backward-forward second differences on that stencil. Write g for the supremum of the first covariant differences of B, and b for the supremum of B, there.

Proposition 1.1. *With the exterior derivative convention*

$$(D_W^* D_W B)_\nu = \sum_{\mu \neq \nu} (\nabla_\mu^* \nabla_\mu B_\nu - \nabla_\mu^* \nabla_\nu B_\mu),$$

one has

$$|([D_W^* D_W, \chi] B)_\nu(x)| \leq (d-1)\big[(2 + 4\tau) s_1 g + (1 + \tau^2) s_2 b\big]. \tag{1}$$

For $1 \leq \tau \leq 5/4$, the right side is at most $c_d(s_1 g + s_2 b)$, with $c_d = (2 + 4\tau)(d - 1)$. Thus $c_4 = 18$ for unitary transport and $c_4 \leq 21$ in the indicated complex neighborhood.

Proof. Set $a_\nu = \partial_\nu \chi$, where the scalar forward difference has the same spacing. The exact product rules give

$$[\nabla_\nu, \chi] = a_\nu T_\nu, \qquad [\nabla_\mu^*, \chi] = -(\partial_\mu^- \chi) T_\mu^{-1},$$
$$\nabla_\mu^*(a_\nu T_\nu B) = a_\nu \nabla_\mu^* T_\nu B - (\partial_\mu^- a_\nu) T_\mu^{-1} T_\nu B.$$

The useful algebraic identity is

$$\nabla_\mu^* T_\nu B = T_\mu^{-1} \nabla_\nu B - T_\mu^{-1} \nabla_\mu B - \nabla_\nu B. \tag{2}$$

Multiplication by η verifies this directly using $T_\mu^{-1} T_\mu = I$. In particular, no two different covariant shifts are commuted. Consequently

$$\begin{aligned}[\nabla_\mu^* \nabla_\nu, \chi] B = a_\nu (T_\mu^{-1} \nabla_\nu B - T_\mu^{-1} \nabla_\mu B - \nabla_\nu B) \\ - (\partial_\mu^- a_\nu) T_\mu^{-1} T_\nu B - (\partial_\mu^- \chi) T_\mu^{-1} \nabla_\nu B,\end{aligned}$$

which is bounded by $(1 + 3\tau) s_1 g + \tau^2 s_2 b$. For $\mu = \nu$, instead use $\nabla_\mu^* T_\mu = -\nabla_\mu$; the bound becomes $(1 + \tau) s_1 g + s_2 b$. Summing the two bounds for each $\mu \neq \nu$ proves (1). All field arguments lie in the radius-η ℓ^∞ stencil. □

For the specific operator $\Delta_{\mathrm{cov}} = \sum_\mu \nabla_\mu^* \nabla_\mu$, the same proof gives

$$|[\Delta_{\mathrm{cov}}, \chi] B| \leq d\big[(1 + \tau) s_1 g + s_2 b\big]. \tag{3}$$

Application to any differently defined Laplacian must include its additional lower-order terms.

1.1 The scale factors in the deposited cutoff

For $W = e^{i\eta a} U$ with U unitary, the invariant Hilbert–Schmidt norm on the Lie algebra gives

$$\max\{\|\mathrm{Ad}_W\|, \left\|\mathrm{Ad}_W^{-1}\right\|\} \leq e^{2\eta \|a\|_{\mathrm{op}}}.$$

Thus $\eta \sup \|a\|_{\mathrm{op}} \leq 1/10$ implies $\tau < 5/4$. If $|a|$ denotes the deposit's normalized Hilbert–Schmidt norm for $N \times N$ matrices, use $\|a\|_{\mathrm{op}} \leq \sqrt{N}|a|$ and the sufficient condition $\eta\sqrt{N} \sup |a| \leq 1/10$. The cutoff in Raw I, lines 111522–111529, is

$$\chi = t + s\widehat{\chi}_\square, \qquad \widehat{\chi}_\square(x) = \prod_\mu \zeta\left(\frac{x_\mu - y_{\square,\mu}}{M \ell_\square}\right), \quad 0 \leq \zeta \leq 1, \quad |\zeta'|, |\zeta''| \leq 5.$$

Integral formulas for finite differences imply $s_1 \leq 5|s|/(M\ell_\square)$ and $s_2 \leq 25|s|/(M^2 \ell_\square^2)$. Set

$$N_y(B) = \max\left\{ \ell_y \sup_{\widetilde{\Delta}(y)} |B|, \ell_y^2 \sup_{\widetilde{\Delta}(y)} |\nabla_W B| \right\}.$$

Assume the stencil is contained in $\widetilde{\Delta}(y)$ and the cover has the stated comparison $\widetilde{\Delta}(y) \cap \square \neq \varnothing \Rightarrow \ell_y \leq L\ell_\square$. For $M \geq 2L^2$, $L \geq 3$,

$$\frac{5\ell_y}{M\ell_\square} + \frac{25\ell_y^2}{M^2 \ell_\square^2} \leq \frac{55}{36} < 2.$$

The cutoff differences vanish when the enlarged stencil misses its support. Hence

$$\ell_y^3 |[D_W^* D_W, \chi] B| \leq 2c_d |s| N_y(B) \mathbf{1}_{\{\widetilde{\Delta}(y) \cap \square \neq \varnothing\}}. \tag{4}$$

The corresponding constant from (3) is $3d$. These constants are independent of lattice spacing, volume, and number of scales. The geometric comparison is checked in Raw I, lines 111630–111683; it is an explicit hypothesis, not a field estimate. Equations (2)–(4) fill the local coefficient-count step marked incomplete at lines 111615–111628.

2 What the combined-current cancellation accomplishes

Use the actual notation $L_\sigma = \Delta_W - P_1(\sigma) + \mathsf{K}$, $Y = H(\sigma)X$, and

$$s_\sigma = L_\sigma A' - \mathfrak{m}_H(\sigma)X + F(W, \mathbb{H}).$$

The definition of the cut current in Raw I, lines 111542–111543, yields the exact identity

$$\begin{aligned} J^\sharp[\chi] = \chi s_\sigma &+ [\Delta_W, \chi]A' - [P_1(\sigma), \chi]A' + [\mathsf{K}, \chi]A' \\ &- [D_W^* D_W, \chi]Y + F(W, \chi\mathbb{H}) - \chi F(W, \mathbb{H}). \end{aligned} \tag{5}$$

The potentially troublesome term $\mathfrak{m}_H X$ cancels. Also $Y = A' - \mathbb{H}$, so its weak first-order norm follows from the other two fields. Proposition 1.1 handles the displayed second-order exterior-derivative commutator without separate Laplacian estimates for these fields.

The projection term remains nonlocal. Its kernel is exactly $P_1(\sigma; x, y)(\chi(y) - \chi(x))$. If the actual decoupled projection has the required $\mathcal{K}_{-2}$ bound with constant C, its commutator has the same bound with constant $2|s|C$ by applying the block estimate to the two products separately. In the inherited kernel algebra this gives

$$|[P_1(\sigma), \chi]A'|_{(-3),\rho} \le 2|s|CL^{d+4}c_1|A'|_{(-1),\rho}.$$

For the sharper spatial profile, retain $\chi(y) - \chi(x)$ instead of its supremum. The required differentiated projection bound for general complex σ is an additional operator input.

2.1 The actual carrier requirements

The deposited transcription of CMP99b(3.37) is $N(a) = \max\{|a|_{(-1)}, |\nabla_U a|_{(-2)}\}$; it contains no Laplacian. The basic carrier adds a small independent current, also without a relation $J = J(W)$. The later increment hypothesis likewise asks only for a first-order potential increment and a combined current increment (Raw I, lines 111779–111783 and 112378–112408). Thus separate second-order bounds are stronger than this immediate consumer requires.

This does not dispose of the entire domain problem. The stored domains also impose curvature and current bounds on derived maps at every retained triple. In particular, their fourth clause concerns $J_{p,X}(M \bullet \mathbb{U})$, not the independent input current. See Raw I, lines 116925–116934 and 119480–119484. A replacement theorem must propagate these *derived combined currents* as well.

2.2 The exact defect at complex decoupling parameters

The earlier chart provides a useful candidate representation but must not be identified with the later global layer by notation alone. Put $D_0 = \Delta_a - Q^* a Q$, with $a = \ell^{-2}$ the scale multiplier. For its forced two-input solution, write

$$P = H_{0,\sigma} b + G_\sigma Y, \qquad Y = \Theta_\sigma H_{0,\sigma} b - f - V_\sigma'(P), \qquad Z = \Theta_\sigma P - f - V_\sigma'(P).$$

Then $Z = (I + \Theta_\sigma G_\sigma)Y$. Define

$$\begin{aligned} E_{H,\sigma} &= D_0 H_{0,\sigma} - Q^*(q_{0,\sigma} - a)\ell^{-1}, \\ E_{G,\sigma} &= D_0 G_\sigma - P_{0,\sigma}^T (I + \Theta_\sigma G_\sigma). \end{aligned}$$

Direct substitution proves

$$D_0 P = P_{0,\sigma}^T Z + Q^*(q_{0,\sigma} - a)\ell^{-1} b + E_{H,\sigma} b + E_{G,\sigma} Y. \tag{6}$$

Both defects vanish at $\sigma = 1$ by the undeformed inverse and block constraint identities. The specified walk decoration at complex σ does not assert those identities.

More explicitly, set $X = D^J_\sigma(P)$, $H = P - H^J_\sigma X$, and $B_\sigma X = Q^*\mathsf{k}_\sigma X + \Lambda^T_\sigma \mathsf{j}_J(H^J_\sigma X)$. Define

$$J_{\mathrm{sub}} = P^T_{0,\sigma} Z + Q^*(q_{0,\sigma} - a)\ell^{-1} b - B_\sigma X - \Delta' H + F(W, H),$$
$$J_{\mathrm{diff}} = (D_0 - \Delta')P - (B_\sigma - \Delta' H^J_\sigma)X + F(W, H).$$

Equation (6) gives the precise discrepancy

$$J_{\mathrm{diff}} - J_{\mathrm{sub}} = E_{H,\sigma} b + E_{G,\sigma} Y. \tag{7}$$

Matching the global operators would add the differences between L_σ and $D_0 - \Delta'$, and between $\mathfrak{m}_H(\sigma)$ and $B_\sigma - \Delta' H^J_\sigma$. These matches are not proved here. An ordinary weak walk estimate does not bound the differentiated defects uniformly as $\eta \downarrow 0$. A local parametrix identity or differentiated walk estimate is the remaining concrete task.

2.3 A possible replacement for the derived-current bound

There is a separate first-derivative route for a configuration which is actually critical under a constraint. It must not be applied to an arbitrary independent-current pair.

Proposition 2.1. *Let Q_s be the linearized constraint at a configuration U, with right inverse E_s on its actual constraint space, $Q_s E_s = I$. Use the bilinear pairings, lattice measures, and transposes of that constraint. Suppose the physical current has the Wilson form $J(U) = D^T_U F_U$, and constrained criticality gives $J(U) = Q^T_s \lambda$. Then*

$$J(U) = \mathcal{R}_s(U) F_U, \qquad \mathcal{R}_s(U) = Q^T_s (D_U E_s)^T. \tag{8}$$

If instead $J(U) = Q^T_s \lambda + r$, the exact additional term is $(I - Q^T_s E^T_s) r$.

Proof. Multiply the criticality equation by E^T_s and use $E^T_s Q^T_s = I$. Substitute $J = D^T_U F_U$, then multiply by Q^T_s. With a residual, the same substitution gives the stated correction. □

For the earlier chart, $QH_0 = \ell^{-1}$, so the correctly normalized candidate is $E_s = H_{0,s}\ell_s$, provided this is the actual composite constraint. This identity uses a first derivative of E_s. For example, suppose blocks in the retained structure have side h_y and metric d_s, and for f supported in block z,

$$\sup_{\Delta_s(y)} |\mathcal{R}_s f| \le C_R h_y^{-1} e^{-\delta d_s(y,z)} \sup |f|.$$

If $|F_U|_{\Delta_s(z)} \le a_z$ and $a_z \le C_a a_y e^{\kappa_s d_s(y,z)}$, with $0 \le \kappa_s < \delta$, then

$$|J(U)|_{\Delta_s(y)} \le C_R C_a h_y^{-1} a_y \sum_z e^{-(\delta - \kappa_s) d_s(y,z)}.$$

Thus a uniform summability bound gives the intended curvature divided by block-side estimate. For the deposited fourth clause, its two scales require $h_y = \ell_{p \wedge \lambda(y)}$ and a curvature profile proportional to $\omega(y)\varepsilon_{m(y)}\ell^{-2}_{m(y)}$. The profile must be controlled in the *retained structure's* metric, not merely in the original hierarchy's metric.

Raw I, lines 128807–128829 and 128923–128930, treats the needed multilevel current estimate as an input. Equation (8) supplies a possible route to that input: identify each composite right inverse and bound its differentiated transpose kernel and profile convolution uniformly. For increments on a fixed structure, one also needs

$$\delta J = \mathcal{R}_s(U_1)\delta F + [\mathcal{R}_s(U_1) - \mathcal{R}_s(U_0)] F_{U_0}.$$

A holomorphic kernel bound on a larger ball would control the second term by Cauchy's formula. Neither that domain nor the uniform composite-kernel statement is proved by the algebraic identity alone.

2.4 Differentiating the first piece of a walk

There is a concrete way to estimate the defects in (6) without assuming they vanish. In the earlier chart put

$$\mathcal{A} = \Delta + DD^*, \quad P_1 = DPD^*, \quad D_0 = \mathcal{A} - P_1.$$

For a local inverse $G_\square$ of $\mathcal{A} - P_{1,\square} + B_\square$, with $B_\square = Q_\square^* a_\square Q_\square$, let $S_\square = h_\square G_\square h_\square$. Assume the local and global differential coefficients agree on the cutoff's required stencil. The local inverse equation gives exactly

$$\begin{aligned} D_0 S_\square = h_\square^2 + [\mathcal{A}, h_\square] G_\square h_\square - P_1 S_\square \\ + h_\square P_{1,\square} G_\square h_\square - h_\square B_\square G_\square h_\square. \end{aligned} \tag{9}$$

This follows simply by moving $\mathcal{A}$ through $h_\square$. The global $Q^* a Q$ term has already canceled in D_0; it is absent from this formula.

Both exterior-derivative parts of the commutator can be handled with first differences. In addition to Proposition 1.1, the scalar-to-bond convention $(DD^*B)_\nu = \sum_\mu \nabla_\nu \nabla_\mu^* B_\mu$ gives

$$\begin{aligned} |([DD^*, \chi]B)_\nu| \le [(1+\tau) + (d-1)(1+\tau+2\tau^2)] s_1 g \\ + [1 + (d-1)\tau^2] s_2 b. \end{aligned}$$

For mixed indices, use $\nabla_\nu T_\mu^{-1} = T_\nu \nabla_\mu^* + \nabla_\nu - \nabla_\mu^*$ and $\nabla_\mu^* = -T_\mu^{-1} \nabla_\mu$ in the product rule. The diagonal term has the bound already proved. Since $\mathcal{A} = D^*D + DD^* + \Delta'$, the remaining local term to bound is $[\Delta', h_\square] G_\square h_\square$. This does not identify the separate global symbol Δ_W with the component Laplacian of Section 1.

For clarity, the summation principle is elementary. Suppose a decorated inverse is an absolutely convergent sum of products $m_\sigma(\gamma) S_{\gamma_0} R_{\gamma_1} \cdots R_{\gamma_n}$. If, in the intended weighted block norms, the sums of the $D_0 S_\square$ majorants from the five terms of (9) and the remainder majorants are bounded by H and r, and $|m_\sigma(\gamma)| \le u^{n+1}$ with $ur < 1$, then

$$\|D_0 G_\sigma\| \le \frac{uH}{1-ur}.$$

Apply D_0 to the first piece, use (9), and sum the geometric series. This requires a bound on those five head terms in the stated norms; it does not infer one from weak convergence.

The relevant head is present in the deposited parametrix at Raw I, lines 25577–25610. After expansion, H_0 and $\widetilde{G}^J$ start with this inverse. Their current insertion Θ_J maps the weak field space boundedly into the forcing/current space. The analogous G_π^J insertion contains $\psi + D\varphi$, so differentiating its head can expose the extra derivative $D\varphi$. Consequently this repair does not automatically establish the previously withheld Laplacian bound for H^J. Its useful target is specifically the defects E_H, E_G, subject to uniform head bounds and the separate identification with the selected global layer.

2.5 The averaging head: measures and a proved flat bound

The last term of (9) needs an actual averaging estimate. Its scale cannot be read from the printed diagonal coefficient while silently changing the transpose convention. Let $F = \eta^d I$ be the fine-lattice measure and $M_c = \mathrm{diag}(\ell_c^d)$ the coarse-volume measure. For the discrete matrix Q, set

$$Q^T_{\mathrm{count}} = F^{-1} Q^{\mathsf{t}}, \qquad Q^T_{\mathrm{vol}} = F^{-1} Q^{\mathsf{t}} M_c.$$

The transpose is bilinear, including the Lie-algebra fiber. Exactly,

$$Q^T_{\mathrm{count}} \,\mathrm{diag}(a_n \ell_n^{d-2}) Q = Q^T_{\mathrm{vol}} \,\mathrm{diag}(a_n \ell_n^{-2}) Q. \tag{10}$$

Raw I, lines 12756–12760, specifies the volume pairing, whereas lines 23961–23968 print the first diagonal coefficient in spaces denoted ℓ^2 without specifying their measures there. Thus (10) is a necessary convention dictionary, not a claim of an inconsistency in the primary paper. If the fine pairing also changes to counting measure, its factor must be converted as well. The dimensionless sequence a_n must retain its own bound.

Proposition 2.2 (Flat bond averaging)**.** *In dimensionless fine coordinates let $M = L^n$ and*

$$(Q_M A)_\mu(c) = M^{-(d+1)} \sum_{x \in B_M(c)} \sum_{t=0}^{M-1} A_\mu(x + te_\mu).$$

Writing $q_M(c,b)$ for its scalar matrix coefficients, one has

$$q_M(c,b) \geq 0, \quad \sum_b q_M(c,b) = 1, \quad q_M(c,b) \leq M^{-d}, \quad \sum_c M^d q_M(c,b) = 1.$$

Consequently, at physical coarse spacing $\ell = M\eta$,

$$\left\| Q_{M\,\mathrm{vol}}^T \, a\ell^{-2} Q_M \right\|_{\infty\to\infty} \leq |a|\ell^{-2}.$$

Restricting the coarse rows or taking a Dirichlet restriction preserves these upper bounds.

Proof. The formula at one step is the flat main term of Raw I, lines 24355–24369; its correction brackets vanish at the identity. Iteration gives the displayed expression: successive block and line coordinates supply independently the base-L digits of x and t. The row sum counts M^{d+1} pairs. For a fixed fine bond, the transverse coordinates select one coarse block. The longitudinal coefficient is $M^{-(d+1)}k_M(s)$, where $k_M(s)$ counts pairs (x_μ, t) in an M-block and $\{0,\dots,M-1\}$ with sum s. Every residue modulo M has exactly M such pairs, so $\sum_z k_M(s - Mz) = M$. This proves the weighted column identity. The final estimate follows by the two positive sums; deleting rows or columns only decreases them. □

Here is a mixed-scale consequence with all needed geometry stated. Suppose at most N_{lev} coarse levels contribute to an output of $hB_\square$, the input and output scales on each averaging support are within a factor R of its coarse scale ℓ_c, and $\sup_n |a_n| \leq a_*$. In the norms $|A|_{(-1)} = \sup_b \ell_b |A_b|$ and $|f|_{(-3)} = \sup_b \ell_b^3 |f_b|$,

$$|hB_\square A|_{(-3)} \leq \|h\|_\infty \, N_{\mathrm{lev}} a_* R^4 |A|_{(-1)}. \tag{11}$$

Indeed $|QA(c)| \leq R\ell_c^{-1}|A|_{(-1)}$, while $\ell_b^3 \ell_c^{-3} \leq R^3$ and the volume-transpose coefficients sum to at most one at each level. The indicated local geometry in Raw I, lines 25742–25747, has only levels $j, j+1$; with the support containment at lines 25479–25483 one may use $N_{\mathrm{lev}} = 2$, $R = L$ there. If the local inverse bound is C_G, this flat head term is at most $2a_* L^4 C_G \, \|h\|_\infty^2$. Conversion to the global hierarchy still requires comparable local and global weights at outer seams. An exponential weight costs at most $e^{\kappa D_Q}$ when each averaging support has diameter D_Q in the retained metric.

For a complex background, a sufficient replacement is the genuinely normalized multilevel kernel estimate

$$\sum_b \|q_n(c,b)\| \leq C_{\mathrm{row}}, \qquad \sup_b \sum_{c \text{ at level } n} (\ell_n/\eta)^d \, \|q_n(c,b)\| \leq C_{\mathrm{col}}, \tag{12}$$

uniformly in n and on the selected complex domain. Then (11) gains only the factor $C_{\mathrm{row}} C_{\mathrm{col}}$. Raw I, lines 46981–46984, explicitly includes L^{-d} in the one-step kernel formula; its later printed bound $|Q_j(U;c,b)| \leq 4$ must therefore not be read as an unweighted matrix bound. The multilevel normalization, support incidence, and complex domain of that source estimate remain to be matched here. A bound obtained by multiplying a fixed constant at every scale would not suffice.

The support count in this proposed completion is elementary. A coarse bond's two endpoint blocks contain $2M^d$ fine sites and at most $2dM^d$ relevant positive fine bonds. A fixed fine bond can be read by at most $2d$ coarse bonds at that level. Consequently a normalized bound $q_n = M^{-d}K_n$, $\|K_n\| \le 4$, would give $C_{\mathrm{row}}, C_{\mathrm{col}} \le 8d$, without multiplying that constant over the hierarchy. This is conditional on the normalized bound on the selected complex domain, not its verification.

There is a second, restricted-patch route to the same primitive. For an operator from spacing s to t, define

$$\|P\|_{\mathrm{av}} = \max\left\{ \sup_c \sum_b \|P_{cb}\|, \quad \sup_b \sum_c (t/s)^d \|P_{cb}\| \right\}.$$

This norm is submultiplicative across successive scales because the volume ratios multiply. The corrected one-step operator has $\|Q(1)\|_{\mathrm{av}} = 1$ by Proposition 2.2. Fixed-dimensional holomorphy on each double block, with a uniform coefficient bound on a fixed small ball, gives by Cauchy's formula

$$\left\| Q(e^{iB}) \right\|_{\mathrm{av}} \le 1 + C_L \|B\|_\infty .$$

If the intermediate backgrounds on the entire read patch satisfy $\|\log \mathcal{M}^p(U)\|_\infty \le C_{\mathrm{av}} b L^{p-n}$ for $0 \le p < n$, within that ball, composition proves

$$\|Q_n(U)\|_{\mathrm{av}} \le \prod_{p=0}^{n-1} (1 + C_L C_{\mathrm{av}} b L^{p-n}) \le \exp\left(\frac{C_L C_{\mathrm{av}} b}{L-1} \right).$$

Every individual coefficient then obeys the normalized bound $\|q_n(c,b)\| \le L^{-nd} \|Q_n(U)\|_{\mathrm{av}}$. The deposited use of CMP98 Proposition 6 at Raw I, lines 17806–17810, has precisely the amplitude-times-block-length form needed for this geometric sequence. Its full primary domain and locality hypotheses have not been verified here. Applying it to the actual head still needs one common regular gauge on the whole read patch, the retained lower bound on local scales, and the local/global weight comparison. The flat norm one belongs to the corrected operator of lines 24355–24365, not to the separate rooted $Q^\flat = Q_{\mathrm{st}} - D_1 a T$ used in Notebook 05.

2.6 The local curvature correction in the head

The stated weak local estimate in Raw I, lines 13617–13620, is

$$|\Delta' A|_{(-3)} \le C_\Delta (a_0 + a_1) |A|_{(-1)}.$$

It implies, without a second derivative of A,

$$|[\Delta', h] A|_{(-3)} \le 2 \|h\|_\infty C_\Delta (a_0 + a_1) |A|_{(-1)}.$$

After composition with $G_\square h$, the bound is $2\|h\|_\infty^2 C_\Delta (a_0 + a_1) C_G$. This gives no inverse-block-size gain; a bounded constant is enough for the head summation principle.

An operator bound is not automatically an absolute kernel-row bound. Here the finite stencil supplies the missing step: testing on one input bounds each weighted matrix coefficient, and summing costs at most the number m_d of stencil bonds. For the indicated one-step locality, the conservative choice $m_d = d5^d$ covers all orientations in a radius-two cube. Exponential weights add their maximal stencil ratio. This proves the required coefficientwise reduction, subject to the same local inverse and retained-weight comparisons. The local P_1 terms, complex averaging estimate, and sum over all heads are still separate inputs to the global current argument.

3 Sharp defects for a complex Gaussian

Let Λ be a finite subset of a countable metric space. Each $i \in \Lambda$ has a real variable $x_i \in \mathbb{R}^D$, with fixed D. Let $C = C^T$ be a complex symmetric covariance on $\mathbb{C}^{D|\Lambda|}$, with

$$\operatorname{Re} C \geq cI, \qquad \|C\| \leq C_*, \qquad c > 0, \tag{13}$$

where real part denotes the Hermitian part. Gaussian integration is over the real coordinate space, with the analytic determinant branch continued from positive real covariances. For fixed measurable functions q_i, assume

$$|q_i(x_i)| \leq \mathbf{1}_{\{|x_i|>R\}}. \tag{14}$$

The sharp cutoff is $q_i = -\mathbf{1}_{\{|x_i|>R\}}$. Its threshold and its real integration domain are fixed as the covariance varies.

Lemma 3.1. *For every nonempty $S \subset \Lambda$,*

$$\left|\mathbb{E}_C \prod_{i\in S} q_i\right| \leq \left[(C_*/c)^D 2^{D/2} \exp\left(-\frac{cR^2}{4C_*^2}\right)\right]^{|S|}.$$

This expectation is holomorphic in C on the accretive cone.

Proof. The marginal covariance C_{SS} inherits both bounds in (13). In $N = D|S|$ coordinates,

$$\operatorname{Re} C_{SS}^{-1} = C_{SS}^{-*}(\operatorname{Re} C_{SS})C_{SS}^{-1} \geq cC_*^{-2}I, \qquad |\det C_{SS}| \geq c^N.$$

The absolute Gaussian density is therefore bounded by $(C_*/c)^N$ times the real isotropic Gaussian of variance C_*^2/c. For a D-dimensional block of variance v, Markov's inequality applied to $e^{|x|^2/(4v)}$ gives $\Pr(|x| > R) \leq 2^{D/2}e^{-R^2/(4v)}$. That comparison Gaussian has independent blocks, proving the bound without independence of the original variables. Locally uniform Gaussian domination proves holomorphy under the integral. No derivative of q_i is taken. □

Lemma 3.2 (A useful precision formulation)**.** *If $A = A^T$, $\operatorname{Re} A \geq aI$, $\|A\| \leq B$, and $C = A^{-1}$, then $\operatorname{Re} C \geq aB^{-2}I$ and $\|C\| \leq a^{-1}$. Moreover,*

$$\left|\mathbb{E}_{A^{-1}} \prod_{i\in S} q_i\right| \leq \left[(B/a)^D 2^{D/2} e^{-aR^2/4}\right]^{|S|}.$$

Proof. Let $P = S$, $Q = \Lambda \setminus S$, and let $S_A = A_{PP} - A_{PQ}A_{QQ}^{-1}A_{QP} = (C_{PP})^{-1}$. For $v = (u, -A_{QQ}^{-1}A_{QP}u)$, one has $Av = (S_A u, 0)$. Hence $\operatorname{Re}\langle u, S_A u\rangle \geq a\|u\|^2$. The lower bound on $\operatorname{Re} C_{PP}$ also gives $\|S_A\| \leq B^2/a$. The absolute marginal density is bounded by $(B/a)^{D|S|}$ times the real Gaussian of precision aI. Apply the same block tail bound. □

4 From marked defects to connected polymer activities

For $i \neq j$ put $a_{ij} = \|C_{ij}\|$ and $a_{ii} = 0$. Fix $\kappa \geq 0$ and assume

$$A_\kappa = \sup_i \sum_{j\neq i} a_{ij} e^{\kappa d(i,j)} < \infty. \tag{15}$$

Let $\ell(S)$ denote the minimum spanning-tree length on S, with $\ell(\{i\}) = 0$. Define safe constants

$$c' = c/2, \quad C' = C_* + c/2, \quad \eta_0 = c/4, \quad K = \frac{2e^2}{\eta_0} = \frac{8e^2}{c}, \quad t_R = (C'/c')^D 2^{D/2} e^{-c'R^2/(4C'^2)}. \tag{16}$$

The symbol η_0 here is a covariance perturbation radius, unrelated to lattice spacing in Sections 1–2.

Theorem 4.1. *There are activities $\zeta(S)$ for all nonempty $S \subset \Lambda$ such that*

$$Z_\Lambda = \mathbb{E}_C \prod_{i\in\Lambda}(1+q_i) = \sum_{\{S_1,\dots,S_m\}\ pairwise\ disjoint} \prod_{j=1}^{m} \zeta(S_j), \tag{17}$$

where the empty family contributes one. For $|S| = n \geq 2$,

$$|\zeta(S)| \leq t_R^n K^{n-1} \sum_{T\ tree\ on\ S} \prod_{i\in S} d_i(T)! \prod_{\{i,j\}\in T} a_{ij}, \tag{18}$$

and $|\zeta(\{i\})| \leq t_R$. For every $\alpha > 0$, set

$$r = \frac{27}{4} e^\alpha t_R K A_\kappa.$$

If $r < 1$, then

$$\sup_x \sum_{S\ni x} |\zeta(S)| e^{\alpha|S|+\kappa\ell(S)} \leq \frac{e^\alpha t_R}{1-r}. \tag{19}$$

All activities are holomorphic in any background on which C is holomorphic and the hypotheses hold uniformly.

Proof. For a marked set S, apply the BKAR forest formula to $F_S(s) = \mathbb{E}_{C(s)} \prod_{i\in S} q_i$, where $C(s)_{ij} = s_{ij}C_{ij}$ for $i \neq j$ and the diagonal is unchanged. The formula is needed only on the convex hull of partition masks [1, 2]; pinching by each partition preserves $\operatorname{Re} C \geq cI$ and contracts operator norm. Thus every interpolated covariance has the same accretivity and norm bounds. No assumption about arbitrary masks in the full parameter cube is needed.

For a tree T, its interpolated mask has entry equal to the minimum edge parameter along the unique tree path. Write this mask as $s^T(u)$. The tree activity is

$$\zeta(S) = \sum_{T\ \text{tree on}\ S} \int_{[0,1]^T} \partial_T F_S(s^T(u))\, du. \tag{20}$$

For a singleton it is the one-block expectation. Forest components factor because their interpolated cross covariances vanish. Therefore $\mathbb{E}_C \prod_{i\in S} q_i$ is the sum over partitions of S of the products of these activities. Expanding $\prod(1+q_i)$ proves (17).

We now bound the derivatives in (20) without differentiating any indicator. Let $d_i = d_i(T)$ and, for each nonzero edge block, use the complex parameter radius

$$\rho_{ij} = \frac{\eta_0}{(d_i+d_j)a_{ij}}.$$

A simultaneous perturbation in these edge parameters changes each block row sum by at most $\eta_0 \sum_{j:\{i,j\}\in T}(d_i+d_j)^{-1} \leq \eta_0$. The same holds for columns, so the operator norm of the perturbation is at most η_0. The complete polydisc remains within the safer bounds $\operatorname{Re} C \geq c'I$, $\|C\| \leq C'$. Lemma 3.1 gives $|F_S| \leq t_R^n$ there. Cauchy's formula in the $n-1$ distinct tree variables yields

$$|\partial_T F_S| \leq t_R^n \eta_0^{-(n-1)} \prod_{\{i,j\}\in T} (d_i+d_j)a_{ij}.$$

An edge with $a_{ij} = 0$ has zero derivative and is simply omitted. Since $d_i + d_j \leq 2d_i d_j$, and $d_i^{d_i} \leq e^{d_i} d_i!$, the last product is bounded by $(2e^2)^{n-1} \prod_i d_i! \prod_{\{i,j\}\in T} a_{ij}$. This proves (18).

For completeness, the exact degree-weighted tree sum is

$$\sum_{T \text{ on } [n]} \prod_i d_i! = (n-2)! \binom{3n-3}{n-2}. \tag{21}$$

Indeed, the Prüfer multiplicities $k_i = d_i - 1$ sum to $n-2$. For fixed multiplicities, multiplication of the number of trees by $\prod_i d_i!$ gives $(n-2)! \prod_i (k_i+1)$. Sum using the coefficient of z^{n-2} in $(1-z)^{-2n}$.

Fix the first vertex at x. Replace unordered marked sets by labelled positions with factor $1/(n-1)!$, drop distinctness, and sum leaf positions successively. The weighted kernel sum at each removal is at most A_κ. Since $\ell(S) \le \sum_{e \in T} d(e)$,

$$\sum_{S \ni x,\ |S|=n} |\zeta(S)| e^{\kappa \ell(S)} \le t_R^n (K A_\kappa)^{n-1} \frac{1}{n-1} \binom{3n-3}{n-2}.$$

For $m = n-1$, the binomial is at most $\binom{3m}{m} \le (27/4)^m$; the latter follows by bounding one term in $(1+1/2)^{3m}$. After inserting $e^{\alpha n}$ and summing, the bound is $e^\alpha t_R [1 + \sum_{m \ge 1} r^m/m]$, which is at most (19). □

5 A zero-free logarithm with an anchored bound

Corollary 5.1. *Under Theorem 4.1, suppose*

$$r \le \tfrac{1}{2}, \qquad 2e^\alpha t_R \le \alpha. \tag{22}$$

Then $Z_\Lambda \neq 0$, with a logarithm continued from zero defect activity, and

$$\log Z_\Lambda = \sum_{\varnothing \neq X \subset \Lambda} e_\Lambda(X), \qquad \sup_x \sum_{X \ni x} e^{\kappa \ell(X)} |e_\Lambda(X)| \le 2e^\alpha t_R.$$

The expansion is absolutely convergent and holomorphic uniformly on any background domain with the same constants. The subscript Λ is retained: consistency between different finite-volume covariances has not been assumed.

Proof. Put $\rho(S) = |\zeta(S)| e^{\kappa \ell(S)}$. Polymers are incompatible exactly when they overlap, including self-incompatibility. Equation (19) gives

$$\sum_{T : T \cap S \neq \varnothing} \rho(T) e^{\alpha |T|} \le \alpha |S|.$$

The Kotecký–Preiss criterion, in its pinned-series formulation [3, 4], therefore bounds the absolute cluster sum pinned at S by $e^{\alpha |S|}$.

Define $e_\Lambda(X)$ by collecting terms of the usual logarithmic cluster series whose union is X; repeated polymers in that series are allowed. A connected overlap graph makes the union of spanning trees inside its polymers connected, so $\ell(X) \le \sum_j \ell(S_j)$. For a cluster containing x, bound its indicator by $\sum_j \mathbf{1}_{\{x \in S_j\}}$ and root at that polymer. The factorial in the cluster series then gives

$$\sum_{X \ni x} e^{\kappa \ell(X)} |e_\Lambda(X)| \le \sum_{S \ni x} \rho(S) e^{\alpha |S|} \le 2e^\alpha t_R.$$

Absolute convergence proves the identity with the finite partition function, nonvanishing, and the stated holomorphy. □

Remark 5.2 (Background derivatives). If C is Fréchet holomorphic and these hypotheses hold uniformly on an open background ball of radius r_0, the family $\{e_\Lambda(X)\}$ is holomorphic with values in its anchored weighted sequence space. Cauchy's formula, using radii $r_0/(2\,\|h\|)$ and $r_0/(2\,\|k\|)$, bounds the mixed second derivative at the center in that norm by $8e^\alpha t_R\,\|h\|\,\|k\|\,/r_0^2$. The boundary radii are approached from below. This statement concerns each finite Λ with a common bound; no assertion of an infinite-volume background functional is implicit. This preserves smallness but gives no additional spatial localization in the supports of h and k.

5.1 Logarithmic real cutoffs

For fixed constants and $R(g) = A(\log g^{-2})^p$, $p > 1/2$, one has $t_{R(g)} = C_0 \exp[-c_0(\log g^{-2})^{2p}]$. Consequently, for every fixed $N > 0$ and real q,

$$t_{R(g)} = o\Big(g^N(\log g^{-2})^q\Big) \quad (g \downarrow 0).$$

To check this, divide logarithms by $(\log g^{-2})^{2p}$: the terms of order $\log g^{-2}$ and $\log\log g^{-2}$ vanish. Thus sufficiently small g enforces (22). This conclusion is uniform in scale only if $D, c, C_*, A_\kappa, A, p$ are themselves uniform. It is a numerical consequence, not a proof that the actual background domains have those constants.

6 A direct quadratic bound without background-support activities

For the scalar running-coupling correction, it can suffice to estimate the total Gaussian cutoff Hessian. This is a narrower task than constructing local effective actions for the complete RG induction.

Assume a metric lattice with volume-independent sums $S_\gamma = \sup_x \sum_i e^{-\gamma d(x,i)} < \infty$ for $\gamma > 0$. In addition to (13), assume $\|C_{ij}\| \le K_0 e^{-\mu d(i,j)}$. Put

$$\nu = \mu/2, \qquad 0 < \kappa < \nu, \qquad \rho = \frac{c}{8S_\nu}.$$

Two symmetric covariance perturbations, each with entries bounded by $e^{-\nu d(i,j)}$, have combined operator norm at most $c/4$ when their two complex coefficients have modulus at most ρ. Thus the complete polydisc has the common safe bounds

$$\operatorname{Re} C \ge c/2, \quad \|C\| \le C_* + c/2, \quad \|C_{ij}\| \le (K_0 + 2\rho)e^{-\nu d(i,j)}.$$

Apply Theorem 4.1 with these constants. Its internal Cauchy margin gives the tail factor

$$t_R^* = 2^{D/2}\left(\frac{C_* + 3c/4}{c/4}\right)^D \exp\left[-\frac{(c/4)R^2}{4(C_* + 3c/4)^2}\right].$$

Choose R large enough that the two smallness conditions (22) hold with these safe constants and $\alpha = 1$. The tree weights in (18) can now all be replaced by the fixed envelope $(K_0 + 2\rho)e^{-\nu d(i,j)}$. Their positive logarithmic cluster series defines one common majorant $M(X)$ such that, throughout every such polydisc,

$$|e_\Lambda(X; C)| \le M(X), \qquad \sup_z \sum_{X \ni z} e^{\kappa \ell(X)} M(X) \le M_0 := 2e\, t_R^*. \tag{23}$$

This is a majorant for the sum of suprema, rather than a separate estimate whose supremum might depend on X. The union coefficient $e_\Lambda(X; C)$ reads only C_{XX}.

Theorem 6.1. *Let $C(s,t)$ be a holomorphic symmetric covariance with value C at the origin. Write $E_x = \partial_s C$, $E_y = \partial_t C$, and $E_{xy} = \partial_s \partial_t C$ there. Suppose, for each nonempty X and $i, j \in X$,*

$$\begin{aligned}
\|(E_x)_{ij}\| &\le B_1 e^{-\mu \operatorname{dist}(x,X)} e^{-\nu d(i,j)}, \\
\|(E_y)_{ij}\| &\le B_1 e^{-\mu \operatorname{dist}(y,X)} e^{-\nu d(i,j)}, \\
\|(E_{xy})_{ij}\| &\le B_2 e^{-\nu[d(x,y)+\operatorname{dist}(x,X)+\operatorname{dist}(y,X)]} e^{-\nu d(i,j)}.
\end{aligned}$$

Then the Gaussian cutoff logarithm $F_\Lambda(C) = \log Z_\Lambda(C)$ satisfies

$$|\partial_s \partial_t F_\Lambda(C(s,t))|_{s=t=0} \le M_0 \left[\frac{B_1^2}{\rho^2} S_{\mu-\kappa} + \frac{B_2}{\rho} S_\nu \right] e^{-\kappa d(x,y)}. \tag{24}$$

Proof. For directions on X with entry envelopes $b_E e^{-\nu d(i,j)}$ and $b_F e^{-\nu d(i,j)}$, extend by zero outside $X \times X$. Cauchy's formula with radii ρ/b_E and ρ/b_F, and the common majorant, gives

$$|De_\Lambda(X)[E]| \le M(X) b_E/\rho, \qquad |D^2 e_\Lambda(X)[E,F]| \le M(X) b_E b_F/\rho^2.$$

Each variable is differentiated once, so no factor two is missing. With $\delta_x = \operatorname{dist}(x,X)$ and similarly for y, the chain rule bounds the summand by

$$M(X) \left[\frac{B_1^2}{\rho^2} e^{-\mu(\delta_x+\delta_y)} + \frac{B_2}{\rho} e^{-\nu[d(x,y)+\delta_x+\delta_y]} \right].$$

The elementary inequality $d(x,y) \le \delta_x + \ell(X) + \delta_y$ implies

$$e^{-\mu(\delta_x+\delta_y)} \le e^{-\kappa d(x,y)} e^{\kappa \ell(X)} e^{-(\mu-\kappa)\delta_x}.$$

For $q > 0$, $e^{-q\delta_x} \le \sum_{z \in X} e^{-q d(x,z)}$. Equation (23) therefore makes the first sum at most $M_0 S_{\mu-\kappa} e^{-\kappa d(x,y)}$ times its coefficient. For the second, use $\nu > \kappa$, drop the y-distance factor, and sum with S_ν. This proves (24). □

6.1 Local precision variations provide the response bounds

Let $A = C^{-1}$ and $A(s,t) = A + sH_x + tH_y$, where each symmetric H_x is supported on $B_{r_s}(x) \times B_{r_s}(x)$, has norm at most H, and the ball has at most n_s sites. Inverse differentiation gives

$$E_x = -CH_xC, \qquad E_{xy} = CH_xCH_yC + CH_yCH_xC.$$

Expanding in blocks yields

$$\begin{aligned}
\|(E_x)_{ij}\| &\le b_1 e^{-\mu[d(i,x)+d(j,x)]}, & b_1 &= n_s^2 H K_0^2 e^{2\mu r_s}, \\
\|(CH_xCH_yC)_{ij}\| &\le b_2 e^{-\mu[d(i,x)+d(x,y)+d(y,j)]}, & b_2 &= n_s^4 H^2 K_0^3 e^{4\mu r_s}.
\end{aligned}$$

The first distance sum is at least both $d(i,j)$ and $2\operatorname{dist}(x,X)$. The second is at least both $d(i,j)$ and $d(x,y) + \delta_x + \delta_y$. Averaging these two lower bounds in each case verifies Theorem 6.1 with $B_1 = b_1$ and $B_2 = 2b_2$. Source supports may overlap, and $x = y$ is allowed.

For a nonlinear precision the exact second response has the additional term $-C(\partial_x \partial_y A)C$. It must be estimated too. For example, if that precision derivative vanishes for $d(x,y) > r_2$ and otherwise has uniformly bounded norm and support in a fixed ball about x, the same first-variation calculation supplies an extra constant times $t_R^* e^{\kappa r_2} e^{-\kappa d(x,y)}$. No locality assumption on first variations alone controls this term.

If a Hessian kernel bounded by $C_H t_R^* e^{-\kappa d(z,w)}$ is contracted with responses $|h_x(z)| \le H_h e^{-m d(x,z)}$, then, for $\lambda = \frac{1}{2}\min(\kappa, m)$, the contraction is bounded by

$$C_H t_R^* H_h^2 S_{m-\lambda} S_{\kappa-\lambda} e^{-\lambda d(x,y)}.$$

This follows by the triangle inequality through x, z, w, y and two geometric sums. Differentiating a nonlinear response chart instead adds its own second-variation term; the displayed contraction alone does not include it.

6.2 Thermodynamic limit and the scalar coefficient

Suppose, additionally, that finite-volume covariances and their first and second source responses converge entrywise on every fixed set, with the preceding bounds uniform. Fixed-set Hessians then converge: the union coefficient has the finite inclusion–exclusion formula

$$e_\Lambda(X; C) = \sum_{Y \subset X} (-1)^{|X \setminus Y|} \log Z_Y(C_{YY}),$$

where the empty-set logarithm is zero. The uniform zero-free domain makes these finite-dimensional functions and their derivatives continuous under the indicated convergence. For sets X not contained in $B_L(x)$, $\operatorname{dist}(x, X) + \ell(X) \ge L$. Spending, for example, $\kappa/2$ of the available tree and source decay gives an omitted Hessian sum bounded by $C t_R^* e^{-\kappa L/2}$, uniformly in volume. Thus the total Hessian converges too. The argument applies to growing tori when fixed balls embed with their ordinary lattice metric and all counting bounds are uniform. Entrywise convergence is an additional hypothesis, not a consequence of the uniform estimate.

At finite volume, any differentiable analytic pointing identity

$$F_{\mathrm{cut},\Lambda}(\xi) - F_{\mathrm{cut},\Lambda}(0) = \sum_{X,z} E_{\mathrm{cut},\Lambda}(X; \xi, z)$$

identifies the total Hessian with the sum of its pointed Hessians. Therefore this direct route can bound the pure Gaussian cutoff part of the deposited quadratic coefficient after its normalized response and infinite-volume kernel have been identified. On $\mathbb{Z}^4$,

$$\sum_y d(x,y)^2 |D_x D_y F_{\mathrm{cut}}| \le C t_R^* \sum_y d(x,y)^2 e^{-\lambda d(x,y)} < \infty.$$

In particular the cutoff contribution to the coupling increment is $O(t_R^*)$ under those identifications.

The exact decomposition is

$$\log \mathbb{E}_C[\chi_R e^V] = \log \mathbb{E}_C[\chi_R] + \log \frac{\mathbb{E}_C[\chi_R e^V]}{\mathbb{E}_C[\chi_R]}.$$

This section bounds the first term. The interaction ratio and the separate Gaussian normalization remain to be controlled. It also does not replace background-local effective actions in the full RG induction. It shows that their complete construction is unnecessary for this particular quadratic estimate once the direct source responses and kernel identification are proved.

7 The remaining Yang–Mills interface

7.1 Matching the actual Gaussian variables

The independent variables are identifiable in the primary construction. CMP109, pp. 266–268, (2.9)–(2.13), linearizes the averaging constraint, rescales by the running coupling, and removes a pivot bond

per next-level bond. The remaining bond vector B is unconstrained; its precision is $\mathcal{C}^*\Delta^{(k)}\mathcal{C}$, where $\mathcal{C}$ denotes the extension map, not a covariance. Each retained bond has $D = \dim_{\mathbb{R}} \mathfrak{g}$ coordinates. The paper explicitly changes the earlier delta gauge to exponential gauge [6].

CMP99b, p. 428 after (3.158), gives a scale-independent positive lower bound for the axial reduced precision; Theorem 3.15, p. 432, gives exponential covariance decay and walks in its stated regular-background setting [5]. These are relevant inputs, but the axial and exponential gauge matrices have different coordinates. Moreover, the primary current is tied to the background, whereas the later analytic carrier permits an independent current.

CMP116, pp. 12–14, (2.1)–(2.10), includes the cutoff-only sector when the interaction sets are empty and the bad-bond set is nonempty. Its localization uses conditioning, exterior whitening, and walks. The complex estimates at pp. 15–17 are structured domination estimates, not an explicitly stated accretive-covariance theorem on every arbitrary complex walk polydisc [7]. Our forest interpolation is a separate, accretivity-preserving device.

The deposit already offers conditional operator constructions here. Raw I, lines 5910–6077 and 7171–7298, derives accretivity and localized covariance bounds from named form-walk and positivity inputs. Lines 5500–5517 distinguish primary tied-current estimates from the free-current hypotheses. Thus the missing task is to verify the hypotheses and coordinate identification, not to pretend that no proposed operator construction exists.

Proposition 7.1 (A sufficient uniform perturbation criterion)**.** *For $\mu \geq 0$, let $\|\cdot\|_\mu$ be the maximum of the exponentially weighted block row and column sums. Suppose the real reference precision A_0 is symmetric, $A_0 \geq 2aI$, $\|A_0\| \leq B_0$, and $\left\|A_0^{-1}\right\|_\mu \leq K_0$. Let the symmetric holomorphic perturbation $V(\xi)$ on a fixed background/current ball satisfy*

$$\|V(\xi)\|_\mu \leq \delta, \qquad \delta \leq a, \qquad K_0\delta < 1.$$

Then $A = A_0 + V$ has $\operatorname{Re} A \geq aI$, $\|A\| \leq B_0 + \delta$, and

$$\left\|A^{-1}\right\|_\mu \leq \frac{K_0}{1 - K_0\delta}.$$

In particular the covariance hypotheses of Theorem 4.1 hold with constants independent of scale whenever the displayed inputs are independent of scale.

Proof. The weighted row/column norm is a matrix algebra norm by the triangle inequality for distance, and dominates operator norm by the Schur test. Accretivity and the upper bound follow immediately. Sum $A^{-1} = (I + A_0^{-1}V)^{-1}A_0^{-1}$ in that algebra. Lemma 3.2 supplies covariance accretivity. □

This criterion reduces the analytic extension to a concrete weighted operator perturbation estimate on the *actual* independent coordinates. It does not prove that estimate or replace background locality. Notebook 06 now proves both real flat precision bounds directly for the explicit full-block Maxwell quadratic model and gives the exact minimizer/coordinate interface to the physical Gaussian. Notebook 07 adds exponential flat kernel decay and an $O(M^{-2})$ refinement limit in weighted kernel norm, uniformly over compatible periodic volumes. This verifies the homogeneous flat reference inputs of the criterion. The weighted bound on the actual complex perturbation is still needed; neither spectral bounds nor spatial decay alone provides it. Notebook 08 additionally proves local fine response legs for the normalized homogeneous flat source minimizer: the potential has uniform physical $C^{1,\alpha}$ bounds, its curl has C^α bounds, and its componentwise Laplacian has a uniform supremum bound, all with exponential coarse-cell decay. The corrected averaging map, scalar gauge projection and normalization are checked explicitly. Notebook 09 now also proves locality of the full flat fine inverse and a common complex neighborhood for the deposited auxiliary current G_π^J. It supplies a quadratic zero-current

precision remainder and all-order localized current derivatives, uniformly in refinement and compatible periodic volume. Notebook 11 also controls the separate nonlinear-constraint term by its corrected fixed-L quadratic composition, and proves a full flat G_1^J current neighborhood in the physical-cell algebra for $M = L^j$. Its passage to the actual retained Gaussian precision and nonflat background domains remain to be matched. Notebook 10 controls small curvature-metric perturbations at fixed Q, which is a further quadratic input and does not identify an entire nonflat Hessian. Notebook 12 additionally proves the full flat current bounds in the homogeneous pointwise N, N_β norms for bounded fine forcing, with no derivative of the current. This is a further matched source input. Its additional heat-kernel argument handles the one-form right-divergence data, with equal exponents for every fixed Hölder exponent strictly between zero and one, and a separate series handles the constrained-current inverse. The scalar member has a separate proof under an explicit uniform positive-penalty hypothesis. Strict dependence on background supports and the retained-coordinate comparison remain distinct obligations. Notebook 13 further straightens the complete nonlinear fixed-reference constraint on a refinement-uniform analytic ball; Notebook 14 computes the actual moving-reference derivative correction, controls the ordinary reference averages, and proves weighted energy-inverse stability for a Wilson operator on a small global physical C^1 ball. Its bare action, physical pairing and left-exponential Hessian chart match the deposited conventions in dimension four. Original-paper verification and the precise graded-curl and effective-Hessian identifications remain separate. Notebook 15 now constructs a nonlinear action minimizer in a homogeneous small-global-field class, with uniform physical $C^{1,\alpha}$ control and analyticity. A gauge flow and an exact ordinary-average transfer remove the fixed-slice and corrected-map restrictions for its local action comparison. Nonflat source pointwise/current domains, mixed retained geometry, and the actual fluctuation integration remain separate interfaces. Notebook 16 further proves weighted responses for the corrected and explicitly chosen ordinary minimizers. Its classical action has derivative equal to the coarse multiplier and a nonlinear Hessian that is a small exponentially local perturbation of the flat Maxwell precision. These estimates concern a classical coarse Hessian, with the ordinary-coordinate convention explicit; they do not identify the precision after quantum integration or give strict background-support activities.

Notebook 17 now proves joint correlated bad-block bounds, including a complex local-event estimate without a total-volume loss. It compares the first-step reduced Wilson Hessian with the flat energy and identifies the homogeneous flat retained Gaussian at all earlier averaging depths using the nonlinear minimum of Notebook 15. Notebook 18 closes the abstract overlapping-interaction expansion under an explicit weighted coefficient norm; with a small local precision decomposition, its activities also have strict background support. Notebook 19 now identifies the actual physical homogeneous pivot Gaussian and bounds its local measure and gauge factors. Notebook 20 matches all local interaction and precision hypotheses for the bare first-step Wilson small-field integral, including its normalization terms. Notebook 21 proves the support-valued classical action and precision bounds uniformly in earlier refinement, and closes one integration of that classical reference action. Notebook 22 constructs the strict stationary-background support expansion and transports those activities to the next coarse variables with explicit radius and support-weight losses. Notebook 23 proves local gauge covariance of the full correction and its quadratic Ward identity. Notebook 24 proves a conditional inherited fluctuation-interaction estimate, and Notebook 26 meets its support-weight test using fixed auxiliary cells. Notebook 25 supplies a smaller classical gauge representative and an analytic composition-germ continuation. Stable quantum domains and amplitudes, including the marginal coupling flow, remain to be proved for the induction.

7.2 Marked-set decay is not background locality

The preceding activities involve the marked locations in S, but $C_{SS}(W)$ may depend on W arbitrarily far away. Consider, for example,

$$C(W) = (1 + \varepsilon W_{x_*})I, \qquad |\varepsilon W_{x_*}| < \tfrac{1}{4}.$$

It satisfies uniform accretivity and norm bounds and has no off-diagonal covariance at all. Nevertheless, every singleton cutoff expectation depends on the remote background coordinate W_{x_*}. Therefore Theorem 4.1 alone cannot supply a family of activities supported in their claimed background domains.

Notebook 04 now proves a sufficient strict-locality theorem when the precision is a sufficiently small sum of local perturbations of a fixed, uniformly accretive, exponentially decaying baseline. It expands the entire marginal density, including its determinant and inverse, and pays for roots in the added support. Its product-domain, symmetry, finite-volume, and actual-operator interfaces remain explicit.

For the actual Yang–Mills Gaussian, the needed further input is a local walk expansion, or an equivalent analytic decoupling construction, for the *background dependence* of the actual Gaussian. It must control the weighted sums of the domains visited by those walks. A truncation to balls with error e^{-cR} does not automatically pay a polymer norm that penalizes their four-dimensional volume. Source-localization and volume compatibility must be checked in the norm actually used by the RG.

7.3 What has to be matched to the deposited construction

The current proof and the Gaussian proof leave a short list of exact interfaces rather than an unidentified analytic remainder:

1. Prove a weighted bound on the combined global current s_σ, on the same complex domains used by the four-stage induction. If the earlier substituted representation is used, estimate (7) and prove the additional global operator identifications.

2. Propagate the derived-current clauses at every retained triple. Weak membership of the basic carrier alone does not establish these.

3. Identify the actual independent Gaussian coordinates and prove uniform accretivity and decay for their full covariance on the fixed complex, independent-current domain. The forest interpolants then inherit these bounds; accretivity of the older complex walk interpolation is not an extra premise of our Gaussian theorem. A bound only on physical tied currents is insufficient.

4. For the full effective-action induction, localize the Gaussian cutoff logarithm in the background variables with an anchored bound on a fixed normalized complex domain. For the scalar quadratic coefficient alone, Section 6 gives the alternative of direct source-response bounds and local convergence. Control the interaction contribution and separate Gaussian normalization as well. Notebook 05 supplies the flat gauge/coercivity bridge and a complete interaction module for disjoint cells. Notebook 18 extends the integration to overlapping supports and combines it with the local precision construction. Its weighted coefficient hypotheses must still be verified for the actual Yang–Mills integrand.

5. Apply the pointed quadratic extraction in Notebook 01 to that actual local family, with uniform normalized source responses. The extraction lemma does not establish its own analytic hypotheses.

These tasks concern a necessary renormalization estimate. The mixed observable hierarchy, all-order continuum bounds, rotation restoration, identification as Yang–Mills, and the mass gap remain separate obligations. Notebook 02 supplies a fixed-coupling reflection argument; it supplies no uniform ultraviolet estimate for this list.

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